

- [GreyCats Analytics -- AI Features API Documentation](#)
 - [Table of Contents](#)
 - [Authentication](#)
 - [Error Handling](#)
 - [Success Response](#)
 - [Error Response](#)
 - [Standard Error Codes](#)
 - [Rate Limits](#)
 - [1. Train Your Brand](#)
 - [1.1 Scrape Brand Website](#)
 - [1.2 Create / Update Brand Profile](#)
 - [1.3 Get Brand Profile](#)
 - [1.4 Upload Knowledge Base File](#)
 - [1.5 List Knowledge Base Files](#)
 - [1.6 Delete Knowledge Base File](#)
 - [2. AI Report Chatbot](#)
 - [2.1 Create Chat Session](#)
 - [2.2 Send Chat Message \(SSE Streaming\)](#)
 - [2.3 Get Chat History](#)
 - [2.4 List Chat Sessions](#)
 - [2.5 Delete Chat Session](#)
 - [2.6 Generate Report Summary](#)
 - [2.7 Rate Message \(Feedback\)](#)
 - [2.8 Mark Action Applied](#)
 - [2.9 Get AI Analytics](#)
 - [3. AI Content Calendar](#)
 - [3.1 Generate Calendar \(Async\)](#)
 - [3.2 Get Calendar \(Poll Status\)](#)
 - [3.3 List Calendars for Client](#)
 - [3.4 Update Calendar Entry](#)
 - [3.5 Delete Calendar](#)
 - [3.6 Create Entry from Chat](#)
 - [3.7 Convert Entry to Scheduled Post](#)
 - [4. AI Creative Suite](#)
 - [4.1 Generate Captions](#)
 - [4.2 Generate Content \(Articles / Scripts\)](#)
 - [4.3 Generate Image \(Sync / Async\)](#)
 - [4.4 Get Creative Assets](#)
 - [4.5 Delete Creative Asset](#)
 - [Appendix: Complete Route Summary](#)

GreyCats Analytics -- AI Features API Documentation

Table of Contents

1. [Authentication](#)
2. [Error Handling](#)
3. [Rate Limits](#)
4. [Train Your Brand](#)
 - [1.1 Scrape Brand Website](#)
 - [1.2 Create / Update Brand Profile](#)
 - [1.3 Get Brand Profile](#)
 - [1.4 Upload Knowledge Base File](#)
 - [1.5 List Knowledge Base Files](#)
 - [1.6 Delete Knowledge Base File](#)
5. [AI Report Chatbot](#)
 - [2.1 Create Chat Session](#)
 - [2.2 Send Chat Message \(SSE Streaming\)](#)
 - [2.3 Get Chat History](#)
 - [2.4 List Chat Sessions](#)
 - [2.5 Delete Chat Session](#)
 - [2.6 Generate Report Summary](#)
 - [2.7 Rate Message \(Feedback\)](#)
 - [2.8 Mark Action Applied](#)
 - [2.9 Get AI Analytics](#)
6. [AI Content Calendar](#)
 - [3.1 Generate Calendar \(Async\)](#)
 - [3.2 Get Calendar \(Poll Status\)](#)
 - [3.3 List Calendars for Client](#)
 - [3.4 Update Calendar Entry](#)
 - [3.5 Delete Calendar](#)
 - [3.6 Create Entry from Chat](#)
 - [3.7 Convert Entry to Scheduled Post](#)
7. [AI Creative Suite](#)
 - [4.1 Generate Captions](#)
 - [4.2 Generate Content \(Articles / Scripts\)](#)
 - [4.3 Generate Image \(Sync / Async\)](#)
 - [4.4 Get Creative Assets](#)
 - [4.5 Delete Creative Asset](#)

Authentication

All AI endpoints require a valid JWT Bearer token. The token must be included in the **Authorization** header of every request.

Authorization: Bearer <JWT_TOKEN>

The authenticated user's ID is extracted from the token and used to enforce resource ownership. All resources (sessions, calendars, brand profiles, assets) are scoped to the authenticated user.

Obtaining a token: Use the platform's login endpoint (`POST /api/auth/login`). Store the returned token using the key `ANALYTICS_TOKEN_KEY_`.

Error Handling

All endpoints return a consistent JSON envelope:

Success Response

```
{
  "success": true,
  "data": { ... },
  "message": "Optional success message"
}
```

Error Response

```
{
  "success": false,
  "message": "Human-readable error description",
  "code": "MACHINE_READABLE_CODE"
}
```

Standard Error Codes

HTTP Status	Code	Description
400	VALIDATION_ERROR	Missing or invalid request parameters
401	--	Missing or invalid JWT token
404	NOT_FOUND	Resource not found or not owned by user
429	RATE_LIMIT_EXCEEDED	Daily rate limit exceeded for this feature
500	AI_PROVIDER_ERROR	AI provider returned an invalid or empty response
500	--	Internal server error

Rate Limits

Rate limits are enforced per-user per-day (UTC midnight reset) via Redis. If Redis is unavailable, limits fail open (requests are allowed).

Feature	Endpoint	Daily Limit
Chat Messages	POST /api/ai/chat/message	100 messages/user/day
Calendar Generation	POST /api/ai/calendar/generate	10 generations/user/day
Caption Generation	POST /api/ai/creative/caption	50 requests/user/day
Content Generation	POST /api/ai/creative/content	30 requests/user/day
Image Generation	POST /api/ai/creative/image	20 images/user/day

When a rate limit is exceeded, the response includes:

```
{
  "success": false,
  "message": "Daily AI message limit reached. Try again tomorrow.",
  "code": "RATE_LIMIT_EXCEEDED",
  "retryAfter": 14523
}
```

The `retryAfter` field (seconds until UTC midnight) is only returned by the chat message endpoint.

1. Train Your Brand

Brand profile and knowledge base management. Brand context is automatically injected into all AI features (chatbot, calendar, creative) for voice-aligned outputs.

1.1 Scrape Brand Website

Scrapes a website URL and uses AI to extract brand information (name, tagline, voice, audience, etc.). Returns a pre-filled brand profile object that the frontend can present for user review before saving.

Endpoint: POST /api/ai/brand/scrape

Auth: Bearer JWT required

Request Body:

Field	Type	Required	Description
url	string	Yes	Full website URL to scrape (e.g., https://example.com)

Success Response (200):

```
{
  "success": true,
  "data": {
    "websiteUrl": "https://example.com",
    "brandName": "Example Co",
    "tagline": "Building the future",
    "industry": "Technology",
    "brandVoice": "PROFESSIONAL",
    "targetAudience": "B2B decision makers aged 25-45",
    "valueProposition": "AI-powered analytics for modern teams",
  }
}
```

```

    "missionStatement": "To democratize data insights",
    "brandStory": "Founded in 2020...",
    "colorPalette": ["#1A1A2E", "#16213E", "#0F3460", "#E94560"],
    "competitors": ["Competitor A", "Competitor B"],
    "keyProducts": ["Product X", "Product Y"],
    "socialHandles": {
      "instagram": "@example",
      "twitter": "@example",
      "linkedin": "company/example"
    }
  },
  "message": "Brand information extracted from website"
}

```

Error Responses:

Status	Condition
400	<code>url</code> is missing or empty
500	Website unreachable (DNS/timeout) -- returns "Could not reach that website. Please check the URL."
500	AI extraction failure

Example:

```

curl -X POST https://api.greycats.com/api/ai/brand/scrape \
  -H "Authorization: Bearer <TOKEN>" \
  -H "Content-Type: application/json" \
  -d '{"url": "https://example.com"}'

```

Notes:

- Scraping uses Cheerio (server-side HTML parsing) -- JavaScript-rendered content may not be captured.
- The AI extraction step can take 5-15 seconds depending on page size.
- Returns partial data if the website has limited information; missing fields will be `null`.

1.2 Create / Update Brand Profile

Creates or updates (upserts) a brand profile for a client. Uses `clientId` as the unique key -- if a profile already exists for that client, it is updated.

Endpoint: `POST /api/ai/brand/profile`

Auth: Bearer JWT required

Request Body:

Field	Type	Required	Description
<code>clientId</code>	number	Yes	Client ID (must be owned by the authenticated user)
<code>brandName</code>	string	Yes	Brand/company name

Field	Type	Required	Description
websiteUrl	string	No	Brand website URL
tagline	string	No	Brand tagline or slogan
industry	string	No	Industry vertical (e.g., "Technology", "Fashion")
brandVoice	string	No	Voice style. Default: "PROFESSIONAL". Options: "PROFESSIONAL", "CASUAL", "BOLD", "FRIENDLY", "AUTHORITATIVE", etc.
targetAudience	string	No	Description of the target audience
valueProposition	string	No	Core value proposition
missionStatement	string	No	Mission statement
brandStory	string	No	Brand origin/story narrative
colorPalette	string[]	No	Array of hex color codes (e.g., ["#1A1A2E", "#E94560"])
competitors	string[]	No	Array of competitor names
keyProducts	string[]	No	Array of key product/service names
socialHandles	object	No	Map of platform to handle (e.g., {"instagram": "@brand"})
doList	string[]	No	Array of brand "do" guidelines (things to always do)
dontList	string[]	No	Array of brand "don't" guidelines (things to avoid)
additionalContext	string	No	Free-form additional context for AI
logoUrl	string	No	URL or path to brand logo

Success Response (200):

```
{
  "success": true,
  "data": {
    "id": "clxyz123...",
    "userId": 5,
    "clientId": 12,
    "brandName": "Example Co",
    "websiteUrl": "https://example.com",
    "tagline": "Building the future",
    "industry": "Technology",
    "brandVoice": "PROFESSIONAL",
    "targetAudience": "B2B decision makers",
    "valueProposition": "AI-powered analytics",
    "missionStatement": "To democratize data",
    "brandStory": "Founded in 2020...",
    "colorPalette": ["#1A1A2E", "#E94560"],
    "competitors": ["Competitor A"],
    "keyProducts": ["Product X"],
    "socialHandles": {"instagram": "@example"},
    "doList": ["Use data-backed claims", "Be concise"],
    "dontList": ["Never use jargon", "Avoid clickbait"],
    "additionalContext": null,
    "logoUrl": null,
    "createdAt": "2026-05-20T10:00:00.000Z",
    "updatedAt": "2026-05-27T14:30:00.000Z"
  },
}
```

```
{
  "message": "Brand profile saved"
}
```

Error Responses:

Status	Condition
400	<code>clientId</code> or <code>brandName</code> missing
404	Client not found or not owned by user
500	Database error

Example:

```
curl -X POST https://api.greycats.com/api/ai/brand/profile \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "clientId": 12,
  "brandName": "Example Co",
  "industry": "Technology",
  "brandVoice": "PROFESSIONAL",
  "targetAudience": "B2B decision makers aged 25-45",
  "doList": ["Use data-backed claims"],
  "dontList": ["Never use jargon"]
}'
```

Notes:

- This is an upsert operation. If a brand profile exists for the given `clientId`, all provided fields are updated. Fields not included in the request body retain their previous values.
- The `colorPalette`, `competitors`, `keyProducts`, `socialHandles`, `doList`, and `dontList` fields are stored as JSON.

1.3 Get Brand Profile

Retrieves the brand profile for a client, including all associated knowledge base files.

Endpoint: `GET /api/ai/brand/profile/:clientId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>clientId</code>	number	Client ID

Success Response (200):

```
{
  "success": true,
  "data": {
    "id": "clxyz123...",
    "userId": 5,
```

```

"clientId": 12,
"brandName": "Example Co",
"websiteUrl": "https://example.com",
"tagline": "Building the future",
"industry": "Technology",
"brandVoice": "PROFESSIONAL",
"targetAudience": "B2B decision makers",
"valueProposition": "AI-powered analytics",
"missionStatement": null,
"brandStory": null,
"colorPalette": ["#1A1A2E"],
"competitors": [],
"keyProducts": [],
"socialHandles": {},
"doList": [],
"dontList": [],
"additionalContext": null,
"logoUrl": null,
"createdAt": "2026-05-20T10:00:00.000Z",
"updatedAt": "2026-05-27T14:30:00.000Z",
"knowledgeBase": [
  {
    "id": "clxyz456...",
    "brandProfileId": "clxyz123...",
    "type": "guideline",
    "label": "Brand Guidelines 2026",
    "fileName": "brand-guide.pdf",
    "filePath": "/app/uploads/ai/brand/abc123.pdf",
    "fileUrl": "/uploads/ai/brand/abc123.pdf",
    "mimeType": "application/pdf",
    "fileSize": 245000,
    "extractedText": "Our brand voice is...",
    "createdAt": "2026-05-22T08:00:00.000Z"
  }
]
}
}

```

Notes:

- Returns `{ "success": true, "data": null }` if no brand profile exists for the client (not a 404 error).
- Knowledge base files are ordered by `createdAt` descending (newest first).

Example:

```

curl -X GET https://api.greycats.com/api/ai/brand/profile/12 \
-H "Authorization: Bearer <TOKEN>"

```

1.4 Upload Knowledge Base File

Uploads a document to the brand's knowledge base. The file's text content is automatically extracted (supports PDF, DOCX, and plain text) and made available to all AI features as additional brand context.

Endpoint: `POST /api/ai/brand/knowledge`

Auth: Bearer JWT required

Content-Type: `multipart/form-data`

Form Fields:

Field	Type	Required	Description
clientId	number	Yes	Client ID
type	string	Yes	Document type. Must be one of: "guideline", "successful_content", "reference"
label	string	No	Human-readable label for the file
file	File	Yes	The document file (PDF, DOCX, TXT, etc.)

Success Response (200):

```
{
  "success": true,
  "data": {
    "id": "clxyz789...",
    "brandProfileId": "clxyz123...",
    "type": "guideline",
    "label": "Brand Guidelines 2026",
    "fileName": "brand-guide.pdf",
    "filePath": "/app/uploads/ai/brand/abc123.pdf",
    "fileUrl": "/uploads/ai/brand/abc123.pdf",
    "mimeType": "application/pdf",
    "fileSize": 245000,
    "extractedText": "Our brand voice is professional yet approachable...",
    "createdAt": "2026-05-27T14:30:00.000Z"
  },
  "message": "Knowledge base file uploaded"
}
```

Error Responses:

Status	Condition
400	Missing clientId, type, or file
400	Invalid type value (must be guideline, successful_content, or reference)
404	Brand profile not found -- create a brand profile first
500	File processing or database error

Example:

```
curl -X POST https://api.greycats.com/api/ai/brand/knowledge \
-H "Authorization: Bearer <TOKEN>" \
-F "clientId=12" \
-F "type=guideline" \
-F "label=Brand Guidelines 2026" \
-F "file=@/path/to/brand-guide.pdf"
```

Notes:

- A brand profile must exist for the client before uploading knowledge files. Call POST /api/ai/brand/profile first.
- Text extraction uses pdf-parse for PDFs and mammoth for DOCX files.
- The extractedText field contains the full extracted text, which is injected into AI prompts for context.

- Files are stored on disk at `/uploads/ai/brand/`.

1.5 List Knowledge Base Files

Lists all knowledge base files for a client's brand profile.

Endpoint: `GET /api/ai/brand/knowledge/:clientId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>clientId</code>	number	Client ID

Success Response (200):

```
{
  "success": true,
  "data": [
    {
      "id": "clxyz789...",
      "brandProfileId": "clxyz123...",
      "type": "guideline",
      "label": "Brand Guidelines 2026",
      "fileName": "brand-guide.pdf",
      "filePath": "/app/uploads/ai/brand/abc123.pdf",
      "fileUrl": "/uploads/ai/brand/abc123.pdf",
      "mimeType": "application/pdf",
      "fileSize": 245000,
      "extractedText": "Our brand voice is...",
      "createdAt": "2026-05-22T08:00:00.000Z"
    }
  ]
}
```

Notes:

- Returns an empty array `[]` if no brand profile exists or no files have been uploaded.
- Files are ordered by `createdAt` descending.

Example:

```
curl -X GET https://api.greycats.com/api/ai/brand/knowledge/12 \
-H "Authorization: Bearer <TOKEN>"
```

1.6 Delete Knowledge Base File

Deletes a specific knowledge base file. Removes both the database record and the physical file from disk.

Endpoint: `DELETE /api/ai/brand/knowledge/:knowledgeId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
knowledgeId	string	Knowledge base file ID (CUID)

Success Response (200):

```
{
  "success": true,
  "message": "Knowledge base item deleted"
}
```

Error Responses:

Status	Condition
404	Knowledge file not found or not owned by user

Example:

```
curl -X DELETE https://api.greycats.com/api/ai/brand/knowledge/clxyz789abc \
-H "Authorization: Bearer <TOKEN>"
```

2. AI Report Chatbot

Conversational AI assistant that analyzes marketing report data. Supports SSE streaming, action item generation, follow-up suggestions, and message feedback.

2.1 Create Chat Session

Creates a new chat session for a specific report template and date range, or resumes an existing active session for the same template. Builds report context by fetching all metrics for the given date range.

Endpoint: POST /api/ai/chat/session

Auth: Bearer JWT required

Request Body:

Field	Type	Required	Description
templateId	number	Yes	Report template ID
clientId	number	No	Client ID (enables brand context and calendar context injection)
dateFrom	string	Yes	Start date (ISO 8601, e.g., "2026-05-01")
dateTo	string	Yes	End date (ISO 8601, e.g., "2026-05-27")

Success Response (200):

```
{
  "success": true,
  "data": {
    "sessionId": "clxyz-session-id...",
    "templateId": 42,
    "clientId": 12,
    "dateFrom": "2026-05-01T00:00:00.000Z",
    "dateTo": "2026-05-27T00:00:00.000Z",
    "reportName": "Monthly Social Media Report",
    "metricCount": 24,
    "suggestions": [
      "How did Instagram engagement change compared to last month?",
      "Which platform drove the most traffic this period?",
      "What day of the week had the highest reach?",
      "How can we improve our Facebook post frequency?"
    ],
    "messages": [],
    "resumed": false,
    "createdAt": "2026-05-27T14:30:00.000Z"
  }
}
```

Resumed Session Response:

When an active session already exists for the same `templateId`, the existing session is returned with its full message history:

```
{
  "success": true,
  "data": {
    "sessionId": "clxyz-existing-session...",
    "templateId": 42,
    "clientId": 12,
    "dateFrom": "2026-05-01T00:00:00.000Z",
    "dateTo": "2026-05-27T00:00:00.000Z",
    "reportName": "Monthly Social Media Report",
    "metricCount": 24,
    "suggestions": [...],
    "messages": [
      {
        "id": "msg-uuid-1",
        "role": "user",
        "content": "How did we perform this month?",
        "actions": null,
        "createdAt": "2026-05-27T10:00:00.000Z"
      },
      {
        "id": "msg-uuid-2",
        "role": "assistant",
        "content": "Based on the report data...",
        "actions": [
          {
            "platform": "instagram",
            "contentType": "Reel",
            "postIdea": "Behind-the-scenes office tour",
            "caption": "Take a peek behind the curtain...",
            "hashtags": ["#BTS", "#OfficeLife"],
            "bestPostTime": "10:00",
            "dayOfWeek": "Wednesday",
            "dataInsight": "Reels had 3.2x more reach than static posts"
          }
        ]
      }
    ],
    "createdAt": "2026-05-27T10:00:05.000Z"
  }
},
```

```
"resumed": true,  
"createdAt": "2026-05-26T09:00:00.000Z"  
}  
}
```

Error Responses:

Status	Condition
400	Missing <code>templateId</code> , <code>dateFrom</code> , or <code>dateTo</code>
400	Invalid date format

Example:

```
curl -X POST https://api.greycats.com/api/ai/chat/session \  
-H "Authorization: Bearer <TOKEN>" \  
-H "Content-Type: application/json" \  
-d '{  
  "templateId": 42,  
  "clientId": 12,  
  "dateFrom": "2026-05-01",  
  "dateTo": "2026-05-27"  
}'
```

Notes:

- The `suggestions` array contains 4 AI-generated, context-aware starter questions based on the actual report data. Falls back to generic suggestions if AI generation fails.
- The `resumed` boolean indicates whether an existing session was resumed (`true`) or a new one was created (`false`).
- The context snapshot (report data) is always refreshed on session creation/resume, so date range changes are reflected.
- Brand context and calendar context are automatically injected if `clientId` is provided.

2.2 Send Chat Message (SSE Streaming)

Sends a user message and streams the AI assistant's response via Server-Sent Events (SSE). After the response completes, optional action items and follow-up suggestions are appended as SSE events.

Endpoint: `POST /api/ai/chat/message`

Auth: Bearer JWT required

Request Body:

Field	Type	Required	Description
<code>sessionId</code>	string	Yes	Chat session ID (from <code>POST /session</code>)
<code>message</code>	string	Yes	User's message text

Response: SSE stream (`Content-Type: text/event-stream`)

The response is a stream of Server-Sent Events. Each event has a **data** field containing a JSON object with a **type** property.

SSE Event Types:

Type	Description	Payload
token	A text chunk from the AI response	{ "type": "token", "text": "Based on " }
done	Signals the end of the text stream	{ "type": "done" }
actions	Actionable content calendar items extracted from the response	{ "type": "actions", "items": [...] }
followups	Follow-up question suggestions	{ "type": "followups", "items": [...] }

SSE Stream Example:

```
data: {"type":"token","text":"Based on "}

data: {"type":"token","text":"the report data, "}

data: {"type":"token","text":"your Instagram engagement "}

data: {"type":"token","text":"increased by **23%**..." }

data: {"type":"done"}

data: {"type":"actions","items":[{"platform":"instagram","contentType":"Reel","postIdea":"Create a trending audio reel showcasing product features","caption":"Ever wondered how we build the future? Here's a sneak peek...", "hashtags":["#Innovation", "#TechLife", "#BTS"], "bestPostTime":"10:00", "dayOfWeek":"Wednesday", "dataInsight": "Reels showed 3.2x higher reach than static posts this period"}]}

data: {"type":"followups","items":["What specific content types drove the most engagement?", "How does our LinkedIn performance compare to Instagram?", "What should we change about our posting schedule?"]}]}
```

Action Item Schema:

Each action in the **actions** array has this shape:

Field	Type	Description
platform	string	"instagram" "facebook" "linkedin" "twitter"
contentType	string	"Post" "Reel" "Story" "Carousel" "Article"
postIdea	string	Specific content idea
caption	string	Draft caption
hashtags	string[]	Relevant hashtags
bestPostTime	string	Recommended post time ("HH:MM" format)
dayOfWeek	string	Recommended day ("Monday" through "Sunday")
dataInsight	string	Data-backed reason why this action is recommended

Error Responses (non-streaming, returned as JSON):

Status	Condition
400	Missing <code>sessionId</code> or <code>message</code>
404	Session not found, not owned by user, or already closed
429	Daily rate limit exceeded (100 messages/day). Includes <code>retryAfter</code> (seconds until midnight).

Example:

```
curl -N -X POST https://api.greycats.com/api/ai/chat/message \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "sessionId": "clxyz-session-id...",
  "message": "How can we improve our Instagram engagement?"
}'
```

Notes:

- The connection uses SSE (`text/event-stream`). Use `EventSource` or a streaming fetch client to consume it.
- The last 10 messages of conversation history are included in each AI call for context continuity.
- Action items are generated via a separate AI call only when the user's question contains action-triggering keywords (e.g., "improve", "strategy", "how can I") AND the AI response contains substantive advice (200+ characters with 2+ advice indicators).
- Actions are de-duplicated against existing calendar entries to avoid suggesting already-planned content.
- Follow-up suggestions are 3 short, context-specific questions generated after each response.
- If the context snapshot is older than 10 minutes, it is automatically refreshed from the database before the AI call.
- The session title is auto-generated from the first user message (async, non-blocking).
- Brand context and calendar entries are injected into the system prompt if the session has a `clientId`.

2.3 Get Chat History

Retrieves all messages for a specific chat session.

Endpoint: `GET /api/ai/chat/session/:sessionId/messages`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>sessionId</code>	string	Chat session ID

Success Response (200):

```
{
  "success": true,
  "data": {
    "session": {
      "id": "clxyz-session-id...",
      "templateId": 42,
      "clientId": 12,
      "dateFrom": "2026-05-01T00:00:00.000Z",
      "dateTo": "2026-05-27T00:00:00.000Z",
      "status": "ACTIVE",
      "createdAt": "2026-05-27T14:30:00.000Z"
    },
    "messages": [
      {
        "id": "msg-uuid-1",
        "role": "user",
        "content": "How did we perform this month?",
        "createdAt": "2026-05-27T14:31:00.000Z"
      },
      {
        "id": "msg-uuid-2",
        "role": "assistant",
        "content": "Based on the report data, your overall performance...",
        "createdAt": "2026-05-27T14:31:05.000Z"
      }
    ]
  }
}
```

Error Responses:

Status	Condition
404	Session not found or not owned by user

Example:

```
curl -X GET https://api.greycats.com/api/ai/chat/session/clxyz-session-id/messages \
-H "Authorization: Bearer <TOKEN>"
```

2.4 List Chat Sessions

Lists all chat sessions for the authenticated user with pagination and optional filtering by template.

Endpoint: `GET /api/ai/chat/sessions`

Auth: Bearer JWT required

Query Parameters:

Parameter	Type	Required	Default	Description
<code>templateId</code>	number	No	--	Filter sessions by report template ID
<code>page</code>	number	No	1	Page number (1-indexed)
<code>limit</code>	number	No	20	Items per page

Success Response (200):

```
{
  "success": true,
  "data": {
    "sessions": [
      {
        "id": "clxyz-session-id...",
        "templateId": 42,
        "clientId": 12,
        "dateFrom": "2026-05-01T00:00:00.000Z",
        "dateTo": "2026-05-27T00:00:00.000Z",
        "status": "ACTIVE",
        "createdAt": "2026-05-27T14:30:00.000Z",
        "_count": {
          "messages": 14
        }
      }
    ],
    "total": 8,
    "page": 1,
    "limit": 20
  }
}
```

Example:

```
curl -X GET "https://api.greycats.com/api/ai/chat/sessions?templateId=42&page=1&limit=10" \
-H "Authorization: Bearer <TOKEN>"
```

2.5 Delete Chat Session

Permanently deletes a chat session and all its messages.

Endpoint: **DELETE** `/api/ai/chat/session/:sessionId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>sessionId</code>	string	Chat session ID

Success Response (200):

```
{
  "success": true,
  "message": "Session cleared"
}
```

Error Responses:

Status	Condition
404	Session not found or not owned by user

Example:

```
curl -X DELETE https://api.greycats.com/api/ai/chat/session/clxyz-session-id \
-H "Authorization: Bearer <TOKEN>"
```

Notes:

- This is a destructive operation. All messages in the session are permanently deleted.
- Messages are deleted first, then the session record.

2.6 Generate Report Summary

Generates a brief 2-3 sentence executive summary of a report. This is a non-streaming, synchronous call.

Endpoint: `POST /api/ai/chat/summary`

Auth: Bearer JWT required

Request Body:

Field	Type	Required	Description
<code>templateId</code>	number	Yes	Report template ID
<code>clientId</code>	number	No	Client ID (enables brand voice in summary)
<code>dateFrom</code>	string	Yes	Start date (ISO 8601)
<code>dateTo</code>	string	Yes	End date (ISO 8601)

Success Response (200):

```
{
  "success": true,
  "data": {
    "summary": "Overall performance improved by 15% this month, driven primarily by a 23% increase in Instagram Reels reach. Facebook engagement declined 8% compared to the previous period, suggesting a need to refresh the content strategy on that platform."
  }
}
```

Error Responses:

Status	Condition
400	Missing <code>templateId</code> , <code>dateFrom</code> , or <code>dateTo</code>

Example:

```
curl -X POST https://api.greycats.com/api/ai/chat/summary \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "templateId": 42,
  "clientId": 12,
  "dateFrom": "2026-05-01",
  "dateTo": "2026-05-27"
}'
```

Notes:

- The summary is capped at 300 tokens (~2-3 sentences).
- If a **clientId** is provided and a brand profile exists, the summary is written in the brand's voice.

2.7 Rate Message (Feedback)

Records thumbs-up or thumbs-down feedback on an AI assistant message.

Endpoint: **PUT** `/api/ai/chat/message/:messageId/feedback`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>messageId</code>	string	Message ID (UUID)

Request Body:

Field	Type	Required	Description
<code>rating</code>	number	Yes	<code>1</code> (thumbs up) or <code>-1</code> (thumbs down)

Success Response (200):

```
{
  "success": true,
  "message": "Feedback recorded"
}
```

Error Responses:

Status	Condition
400	<code>rating</code> is not <code>1</code> or <code>-1</code>
404	Message not found or not owned by user

Example:

```
curl -X PUT https://api.greycats.com/api/ai/chat/message/msg-uuid-123/feedback \
-H "Authorization: Bearer <TOKEN>" \
```

```
-H "Content-Type: application/json" \  
-d '{"rating": 1}'
```

2.8 Mark Action Applied

Marks a specific action item (from an AI response) as "applied" -- e.g., when the user adds it to their calendar.

Endpoint: `PUT /api/ai/chat/message/:messageId/action/:actionIndex/apply`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>messageId</code>	string	Message ID (UUID)
<code>actionIndex</code>	number	Zero-based index of the action in the message's <code>actions</code> array

Success Response (200):

```
{  
  "success": true,  
  "message": "Action marked as applied"  
}
```

Error Responses:

Status	Condition
400	Invalid action index (out of range)
404	Message not found or not owned by user

Example:

```
curl -X PUT https://api.greycats.com/api/ai/chat/message/msg-uuid-123/action/0/apply \  
-H "Authorization: Bearer <TOKEN>"
```

Notes:

- Sets `applied: true` on the action object within the message's JSON `actions` field.
- This is an idempotent operation.

2.9 Get AI Analytics

Returns aggregate analytics about the authenticated user's AI chatbot usage over the last 30 days.

Endpoint: `GET /api/ai/chat/analytics`

Auth: Bearer JWT required

Success Response (200):

```
{
  "success": true,
  "data": {
    "totalSessions": 25,
    "activeSessions": 3,
    "totalMessages": 142,
    "avgMessagesPerSession": 6,
    "positiveRatings": 45,
    "negativeRatings": 3,
    "topQuestions": [
      { "question": "how can we improve instagram engagement?", "count": 4 },
      { "question": "what were the key highlights this month?", "count": 3 }
    ],
    "dailyChart": [
      { "date": "2026-05-01", "count": 5 },
      { "date": "2026-05-02", "count": 8 },
      { "date": "2026-05-03", "count": 3 }
    ]
  }
}
```

Response Fields:

Field	Type	Description
totalSessions	number	Total chat sessions (all time)
activeSessions	number	Currently active (non-deleted) sessions
totalMessages	number	Total messages across all sessions
avgMessagesPerSession	number	Rounded average messages per session
positiveRatings	number	Count of thumbs-up ratings
negativeRatings	number	Count of thumbs-down ratings
topQuestions	array	Top 10 most frequent user questions (last 30 days), normalized and truncated to 100 chars
dailyChart	array	Daily message counts (last 30 days), sorted by date ascending

Example:

```
curl -X GET https://api.greycats.com/api/ai/chat/analytics \
-H "Authorization: Bearer <TOKEN>"
```

3. AI Content Calendar

AI-powered content calendar generation with async processing, entry management, and integration with the social media scheduler.

3.1 Generate Calendar (Async)

Starts asynchronous generation of a content calendar for a specific month. Returns immediately with a `calendarId` and `GENERATING` status. The client should poll `GET /api/ai/calendar/:calendarId` for progress.

Endpoint: `POST /api/ai/calendar/generate`

Auth: Bearer JWT required

Rate Limit: 10 generations/user/day

Request Body:

Field	Type	Required	Default	Description
<code>clientId</code>	number	Yes	--	Client ID
<code>month</code>	number	Yes	--	Month (1-12)
<code>year</code>	number	Yes	--	Year (2024-2030)
<code>platforms</code>	string[]	No	<code>["instagram", "facebook"]</code>	Platforms to generate content for. Valid: <code>"instagram"</code> , <code>"facebook"</code> , <code>"linkedin"</code> , <code>"twitter"</code>
<code>goals</code>	string	No	<code>null</code>	Content goals description (e.g., "Increase brand awareness and drive website traffic")
<code>postsPerWeek</code>	number	No	<code>5</code>	Number of posts per week (1-14)

Success Response (200):

```
{
  "success": true,
  "data": {
    "calendarId": "clxyz-calendar-id...",
    "status": "GENERATING",
    "message": "Calendar generation started. Poll /api/ai/calendar/clxyz-calendar-id... for status."
  }
}
```

Error Responses:

Status	Condition
400	Missing <code>clientId</code> , <code>month</code> , or <code>year</code>
400	<code>month</code> not 1-12 or <code>year</code> not 2024-2030
400	No valid platforms provided
404	Client not found or not owned by user
429	Rate limit exceeded (10/day)

Example:

```
curl -X POST https://api.greycats.com/api/ai/calendar/generate \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "clientId": 12,
  "month": 6,
  "year": 2026,
  "platforms": ["instagram", "facebook", "linkedin"],
  "goals": "Increase brand awareness and drive website traffic",
  "postsPerWeek": 7
}'
```

Notes:

- Generation happens asynchronously. The progress goes through stages: 10% (started) -> 20% (brand context loaded) -> 30% (AI config + historical data) -> 40% (prompt built) -> 70% (AI response received) -> 85% (response parsed) -> 100% (entries saved).
- If generation fails, the calendar status is set to **"FAILED"** with **progress: 0**.
- Brand context, knowledge base, and historical performance data (best posting times, top-performing content types) are automatically injected into the AI prompt.
- **postsPerWeek** is clamped to the range 1-14.

3.2 Get Calendar (Poll Status)

Retrieves a calendar with its status, progress, and all entries. Use this to poll for async generation progress.

Endpoint: **GET** `/api/ai/calendar/:calendarId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>calendarId</code>	string	Calendar ID

Success Response (200) -- Generating:

```
{
  "success": true,
  "data": {
    "id": "clxyz-calendar-id...",
    "userId": 5,
    "clientId": 12,
    "month": 6,
    "year": 2026,
    "platforms": ["instagram", "facebook", "linkedin"],
    "goals": "Increase brand awareness",
    "postsPerWeek": 7,
    "status": "GENERATING",
    "progress": 40,
    "generatedAt": null,
    "createdAt": "2026-05-27T14:30:00.000Z",
    "updatedAt": "2026-05-27T14:30:05.000Z",
    "entries": []
  }
}
```

```
}  
}
```

Success Response (200) -- Completed:

```
{  
  "success": true,  
  "data": {  
    "id": "clxyz-calendar-id...",  
    "userId": 5,  
    "clientId": 12,  
    "month": 6,  
    "year": 2026,  
    "platforms": ["instagram", "facebook", "linkedin"],  
    "goals": "Increase brand awareness",  
    "postsPerWeek": 7,  
    "status": "COMPLETED",  
    "progress": 100,  
    "generatedAt": "2026-05-27T14:31:00.000Z",  
    "createdAt": "2026-05-27T14:30:00.000Z",  
    "updatedAt": "2026-05-27T14:31:00.000Z",  
    "entries": [  
      {  
        "id": "clxyz-entry-1...",  
        "calendarId": "clxyz-calendar-id...",  
        "date": "2026-06-02",  
        "dayOfWeek": "Monday",  
        "platform": "instagram",  
        "contentType": "Reel",  
        "postIdea": "Behind-the-scenes look at product development",  
        "caption": "Ever wondered how the magic happens? Take a peek behind the curtain...",  
        "hashtags": ["#BTS", "#ProductDev", "#Innovation"],  
        "bestPostTime": "10:00",  
        "dataInsight": "Reels posted on Mondays at 10 AM had 2.5x more reach",  
        "imageUrl": null,  
        "source": null,  
        "status": "DRAFT",  
        "createdAt": "2026-05-27T14:31:00.000Z",  
        "updatedAt": "2026-05-27T14:31:00.000Z"  
      }  
    ]  
  }  
}
```

Calendar Status Values:

Status	Description
GENERATING	Calendar is being generated. Check progress (0-100).
COMPLETED	Generation finished. Entries are available.
FAILED	Generation failed. progress is reset to 0.

Error Responses:

Status	Condition
404	Calendar not found or not owned by user

Example:


```
curl -X GET https://api.greycats.com/api/ai/calendar/clxyz-calendar-id \
-H "Authorization: Bearer <TOKEN>"
```

Notes:

- Entries are sorted by **date** ascending, then **bestPostTime** ascending.
- Poll every 2-3 seconds while **status** is "GENERATING". Stop polling when **status** changes to "COMPLETED" or "FAILED".

3.3 List Calendars for Client

Lists all calendars for a specific client with pagination.

Endpoint: **GET** `/api/ai/calendar/client/:clientId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>clientId</code>	number	Client ID

Query Parameters:

Parameter	Type	Required	Default	Description
<code>page</code>	number	No	<code>1</code>	Page number (1-indexed)
<code>limit</code>	number	No	<code>10</code>	Items per page (max 50)

Success Response (200):

```
{
  "success": true,
  "data": [
    {
      "calendarId": "clxyz-calendar-id...",
      "month": 6,
      "year": 2026,
      "status": "COMPLETED",
      "progress": 100,
      "platforms": ["instagram", "facebook", "linkedin"],
      "entryCount": 28,
      "generatedAt": "2026-05-27T14:31:00.000Z",
      "createdAt": "2026-05-27T14:30:00.000Z"
    }
  ]
}
```

Example:

```
curl -X GET "https://api.greycats.com/api/ai/calendar/client/12?page=1&limit=10" \
-H "Authorization: Bearer <TOKEN>"
```

3.4 Update Calendar Entry

Updates fields on an individual calendar entry.

Endpoint: **PUT** `/api/ai/calendar/entry/:entryId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>entryId</code>	string	Calendar entry ID

Request Body (all fields optional):

Field	Type	Description
<code>postIdea</code>	string	Updated post idea
<code>caption</code>	string	Updated caption
<code>bestPostTime</code>	string	Updated post time (" <code>HH:MM</code> " format)
<code>platform</code>	string	Updated platform
<code>contentType</code>	string	Updated content type
<code>hashtags</code>	string[]	Updated hashtags array
<code>imageUrl</code>	string	Image URL (e.g., from AI Creative Suite)
<code>status</code>	string	Entry status: " <code>DRAFT</code> ", " <code>APPROVED</code> ", " <code>PUBLISHED</code> "

Success Response (200):

```
{
  "success": true,
  "data": {
    "id": "clxyz-entry-1...",
    "calendarId": "clxyz-calendar-id...",
    "date": "2026-06-02",
    "dayOfWeek": "Monday",
    "platform": "instagram",
    "contentType": "Reel",
    "postIdea": "Updated: Behind-the-scenes product tour",
    "caption": "Updated caption here...",
    "hashtags": ["#BTS", "#ProductDev"],
    "bestPostTime": "11:00",
    "dataInsight": "Reels posted on Mondays had 2.5x more reach",
    "imageUrl": "/uploads/ai/creative/img-123.png",
    "source": null,
    "status": "APPROVED",
    "createdAt": "2026-05-27T14:31:00.000Z",
    "updatedAt": "2026-05-27T15:00:00.000Z"
  }
}
```

```
}  
}
```

Error Responses:

Status	Condition
404	Entry not found or not owned by user

Example:

```
curl -X PUT https://api.greycats.com/api/ai/calendar/entry/clxyz-entry-1 \  
-H "Authorization: Bearer <TOKEN>" \  
-H "Content-Type: application/json" \  
-d '{  
  "caption": "Updated caption with better CTA",  
  "bestPostTime": "11:00",  
  "status": "APPROVED"  
}
```

Notes:

- Only provided fields are updated; omitted fields remain unchanged.
- Ownership is verified via the calendar's **userId**.

3.5 Delete Calendar

Permanently deletes a calendar and all its entries.

Endpoint: **DELETE** `/api/ai/calendar/:calendarId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
calendarId	string	Calendar ID

Success Response (200):

```
{  
  "success": true,  
  "message": "Calendar deleted"  
}
```

Error Responses:

Status	Condition
404	Calendar not found or not owned by user

Example:

```
curl -X DELETE https://api.greycats.com/api/ai/calendar/clxyz-calendar-id \
-H "Authorization: Bearer <TOKEN>"
```

Notes:

- Cascade deletes all associated entries.

3.6 Create Entry from Chat

Creates a calendar entry from a chatbot action item. If no `calendarId` is provided, automatically finds or creates a calendar for the entry's month.

Endpoint: `POST /api/ai/calendar/entry/from-chat`

Auth: Bearer JWT required

Request Body:

Field	Type	Required	Description
<code>clientId</code>	number	Yes	Client ID
<code>calendarId</code>	string	No	Target calendar ID. If omitted, auto-finds or creates one for the month.
<code>date</code>	string	Yes	Date ("YYYY-MM-DD" format)
<code>dayOfWeek</code>	string	No	Day of week (auto-calculated from date if omitted)
<code>platform</code>	string	Yes	Platform name
<code>contentType</code>	string	No	Content type (default: "Post")
<code>postIdea</code>	string	Yes	Post idea / content description
<code>caption</code>	string	No	Draft caption
<code>hashtags</code>	string[]	No	Array of hashtags
<code>bestPostTime</code>	string	No	Post time ("HH:MM" format)
<code>dataInsight</code>	string	No	Data insight supporting this content

Success Response (200):

```
{
  "success": true,
  "data": {
    "id": "clxyz-entry-new...",
    "calendarId": "clxyz-calendar-id...",
    "date": "2026-06-05",
    "dayOfWeek": "Thursday",
    "platform": "instagram",
    "contentType": "Reel",
    "postIdea": "Behind-the-scenes office tour",
    "caption": "Take a peek behind the curtain...",
    "hashtags": ["#BTS", "#OfficeLife"],
    "bestPostTime": "10:00",
    "dataInsight": "Reels had 3.2x more reach",
```

```
{
  "source": "chat",
  "status": "DRAFT",
  "createdAt": "2026-05-27T15:00:00.000Z",
  "updatedAt": "2026-05-27T15:00:00.000Z"
},
{
  "message": "Entry added to calendar"
}
```

Error Responses:

Status	Condition
400	Missing <code>clientId</code> , <code>date</code> , <code>platform</code> , or <code>postIdea</code>

Example:

```
curl -X POST https://api.greycats.com/api/ai/calendar/entry/from-chat \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "clientId": 12,
  "date": "2026-06-05",
  "platform": "instagram",
  "contentType": "Reel",
  "postIdea": "Behind-the-scenes office tour",
  "caption": "Take a peek behind the curtain...",
  "hashtags": ["#BTS", "#OfficeLife"],
  "bestPostTime": "10:00",
  "dataInsight": "Reels had 3.2x more reach"
}'
```

Notes:

- Entries created from chat are tagged with `source: "chat"` to distinguish them from AI-generated entries.
- If no calendar exists for the month derived from `date`, one is automatically created with `status: "COMPLETED"`.
- `dayOfWeek` is auto-calculated from the `date` if not provided.

3.7 Convert Entry to Scheduled Post

Converts a calendar entry into a draft scheduled post in the Social Media Scheduler. Marks the calendar entry as `"APPROVED"`.

Endpoint: `POST /api/ai/calendar/entry/:entryId/schedule`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
<code>entryId</code>	string	Calendar entry ID

Request Body (all fields optional):

Field	Type	Description
<code>message</code>	string	Override the post message/content. If omitted, uses the entry's caption + hashtags.
<code>date</code>	string	Override the date ("YYYY-MM-DD"). If omitted, uses the entry's date.
<code>time</code>	string	Override the time ("HH:MM"). If omitted, uses the entry's <code>bestPostTime</code> or "09:00".

Success Response (200):

```
{
  "success": true,
  "data": {
    "scheduledPostId": "clxyz-post-id...",
    "scheduledFor": "2026-06-05T10:00:00.000Z"
  },
  "message": "Entry converted to scheduled post"
}
```

Error Responses:

Status	Condition
404	Entry not found or not owned by user

Example:

```
curl -X POST https://api.greycats.com/api/ai/calendar/entry/clxyz-entry-1/schedule \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "date": "2026-06-05",
  "time": "10:00"
}'
```

Notes:

- The scheduled post is created with `status: "PENDING"`.
- Markdown formatting in the message is converted to Unicode bold characters for social media compatibility (e.g., `**bold**` becomes `bold`).
- For articles (caption > 500 characters), a short teaser is auto-generated instead of using the full content.
- The calendar entry's status is automatically updated to `"APPROVED"`.

4. AI Creative Suite

AI-powered creative content generation: captions, articles, scripts, and images.

4.1 Generate Captions

Generates multiple social media caption variants for a given topic and platform.

Endpoint: `POST /api/ai/creative/caption`

Auth: Bearer JWT required

Rate Limit: 50 requests/user/day

Request Body:

Field	Type	Required	Default	Description
<code>clientId</code>	number	Yes	--	Client ID
<code>platform</code>	string	Yes	--	Target platform (e.g., <code>"instagram"</code> , <code>"facebook"</code> , <code>"linkedin"</code> , <code>"twitter"</code>)
<code>topic</code>	string	Yes	--	Content topic or description
<code>goal</code>	string	No	<code>"engagement"</code>	Content goal (e.g., <code>"engagement"</code> , <code>"awareness"</code> , <code>"conversion"</code> , <code>"traffic"</code>)
<code>tone</code>	string	No	Brand voice or <code>"PROFESSIONAL"</code>	Tone override (e.g., <code>"CASUAL"</code> , <code>"BOLD"</code> , <code>"FRIENDLY"</code>)
<code>includeHashtags</code>	boolean	No	<code>true</code>	Whether to include hashtags with each caption
<code>count</code>	number	No	<code>3</code>	Number of captions to generate (max 5)

Success Response (200):

```
{
  "success": true,
  "data": {
    "captions": [
      {
        "id": "caption_1",
        "text": "Ever wondered what sets industry leaders apart? It's not just innovation -- it's consistency. Here's how we're redefining excellence, one step at a time.",
        "hashtags": ["#Leadership", "#Innovation", "#ExampleCo"],
        "characterCount": 152,
        "tone": "PROFESSIONAL"
      },
      {
        "id": "caption_2",
        "text": "3 things we learned from our best-performing campaign this quarter (number 2 will surprise you)...",
        "hashtags": ["#MarketingTips", "#DataDriven"],
        "characterCount": 98,
        "tone": "PROFESSIONAL"
      },
      {
        "id": "caption_3",
        "text": "What does success look like for your brand? We asked 500 marketers and the answers might change how you think about growth.",
        "hashtags": ["#Growth", "#MarketingInsights"],
        "characterCount": 128,
        "tone": "PROFESSIONAL"
      }
    ],
    "assetId": "clxyz-asset-id..."
  }
}
```

```
}  
}
```

Error Responses:

Status	Condition
400	Missing clientId , platform , or topic
404	Client not found or not owned by user
429	Rate limit exceeded (50/day)
500	AI returned invalid or empty captions (AI_PROVIDER_ERROR)

Example:

```
curl -X POST https://api.greycats.com/api/ai/creative/caption \  
-H "Authorization: Bearer <TOKEN>" \  
-H "Content-Type: application/json" \  
-d '{  
  "clientId": 12,  
  "platform": "instagram",  
  "topic": "Summer product launch campaign",  
  "goal": "awareness",  
  "tone": "CASUAL",  
  "includeHashtags": true,  
  "count": 4  
'
```

Notes:

- Each caption is generated with a different approach (question, story, call-to-action, statistic) to provide variety.
- Brand voice from the brand profile is automatically applied. The **tone** field overrides the brand voice.
- Historical performance data (best-performing content types, engagement patterns) is injected into the AI prompt.
- Generated captions are automatically saved as a creative asset. The **assetId** can be used to retrieve them later.
- Character counts are provided for platform-specific length validation.

4.2 Generate Content (Articles / Scripts)

Generates long-form content: blog articles or video scripts.

Endpoint: **POST** **/api/ai/creative/content**

Auth: Bearer JWT required

Rate Limit: 30 requests/user/day

Request Body:

Field	Type	Required	Default	Description
<code>clientId</code>	number	Yes	--	Client ID
<code>topic</code>	string	Yes	--	Content topic or title
<code>contentType</code>	string	No	"article"	Type: "article" or "script"
<code>platform</code>	string	No	"a website" (article) / "Instagram" (script)	Target platform
<code>context</code>	string	No	--	Additional context or instructions for the AI

Success Response (200) -- Article:

```
{
  "success": true,
  "data": {
    "content": "# How AI Is Transforming Marketing Analytics\n\nIn today's data-driven landscape, marketing teams...\n\n## The Rise of Predictive Analytics\n\nPredictive analytics has become...\n\n## Key Takeaways\n\n- **Automation** reduces manual reporting time by 60%\n- **AI insights** uncover patterns humans miss\n- **Real-time data** enables faster decision-making\n\n## Ready to Transform Your Marketing?\n\nStart your journey with GreyCats Analytics today...",
    "contentType": "article",
    "wordCount": 650
  }
}
```

Success Response (200) -- Script:

```
{
  "success": true,
  "data": {
    "content": "[HOOK - 0:00-0:03]\n[VISUAL] Quick montage of marketing dashboards\n[TEXT ON SCREEN] \"Your data is trying to tell you something\"\n\n[MAIN - 0:03-0:25]\n[VISUAL] Screen recording of analytics platform\n[VOICEOVER] \"Most marketers spend hours creating reports...\"\n\n[CTA - 0:25-0:30]\n[VISUAL] Logo + website URL\n[TEXT ON SCREEN] \"Try GreyCats Analytics free\"\n[VOICEOVER] \"Link in bio to get started\"",
    "contentType": "script",
    "wordCount": 85
  }
}
```

Error Responses:

Status	Condition
400	Missing <code>clientId</code> or <code>topic</code>
404	Client not found or not owned by user
429	Rate limit exceeded (30/day)

Example:

```
curl -X POST https://api.greycats.com/api/ai/creative/content \
-H "Authorization: Bearer <TOKEN>" \
```

```
-H "Content-Type: application/json" \
-d '{
  "clientId": 12,
  "topic": "How AI Is Transforming Marketing Analytics",
  "contentType": "article",
  "platform": "linkedin",
  "context": "Focus on SMB use cases and include statistics"
}'
```

Notes:

- Articles are 500-800 words with markdown formatting (headings, bold, bullet points), SEO-friendly, and include a CTA. Max tokens: 4096.
- Scripts include **[VISUAL]** and **[VOICEOVER/TEXT]** markers for scene-by-scene production guidance. Max tokens: 2048.
- Brand context is automatically injected for voice-aligned output.
- Content is not auto-saved as an asset (unlike captions). The frontend should handle saving or display.

4.3 Generate Image (Sync / Async)

Generates an AI image using a two-step process: first generates an optimized image prompt using AI, then creates the image via OpenAI's image generation API (gpt-image-1).

Endpoint: **POST** `/api/ai/creative/image`

Auth: Bearer JWT required

Rate Limit: 20 images/user/day

Request Body:

Field	Type	Required	Default	Description
<code>clientId</code>	number	Yes	--	Client ID
<code>intent</code>	string	Yes	--	Description of what the image should depict
<code>platform</code>	string	No	<code>"instagram"</code>	Target platform (affects aspect ratio optimization)
<code>aspectRatio</code>	string	No	<code>"1:1"</code>	Aspect ratio: <code>"1:1"</code> , <code>"16:9"</code> , <code>"9:16"</code> , <code>"4:5"</code>
<code>style</code>	string	No	<code>"photorealistic"</code>	Image style (e.g., <code>"photorealistic"</code> , <code>"illustration"</code> , <code>"flat design"</code> , <code>"watercolor"</code>)
<code>mode</code>	string	No	<code>"sync"</code>	<code>"sync"</code> (wait for result) or <code>"async"</code> (return immediately, generate in background)

Aspect Ratio to Pixel Size Mapping:

Aspect Ratio	Pixel Size	Best For
1:1	1024x1024	Instagram feed, Facebook
16:9	1536x1024	LinkedIn, YouTube thumbnails
9:16	1024x1536	Instagram Stories, Reels
4:5	1024x1024	Instagram portrait posts

Success Response (200) -- Sync Mode (completed):

```
{
  "success": true,
  "data": {
    "assetId": "clxyz-asset-id...",
    "imageUrl": "/uploads/ai/creative/img-1716825600000.png",
    "promptUsed": "A professional, modern office workspace photographed from a 45-degree angle, featuring a large monitor displaying colorful analytics dashboards with blue and coral accent colors (#1A1A2E, #E94560), natural daylight streaming through floor-to-ceiling windows, minimalist desk accessories, shallow depth of field, photorealistic, 8k quality",
    "aspectRatio": "1:1",
    "generatedAt": "2026-05-27T15:00:00.000Z",
    "status": "COMPLETED"
  }
}
```

Success Response (200) -- Sync Mode (failed):

```
{
  "success": true,
  "data": {
    "assetId": "clxyz-asset-id...",
    "imageUrl": null,
    "promptUsed": "A professional office workspace...",
    "aspectRatio": "1:1",
    "generatedAt": "2026-05-27T15:00:00.000Z",
    "status": "FAILED"
  }
}
```

Success Response (200) -- Async Mode:

```
{
  "success": true,
  "data": {
    "assetId": "clxyz-asset-id...",
    "imageUrl": null,
    "promptUsed": "A professional office workspace...",
    "status": "GENERATING",
    "message": "Image is being generated. Check the gallery in ~30 seconds."
  }
}
```

Error Responses:

Status	Condition
400	Missing <code>clientId</code> or <code>intent</code>

Status	Condition
404	Client not found or not owned by user
429	Rate limit exceeded (20/day)

Example (Sync):

```
curl -X POST https://api.greycats.com/api/ai/creative/image \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "clientId": 12,
  "intent": "Modern office workspace with analytics dashboards on screen",
  "platform": "instagram",
  "aspectRatio": "1:1",
  "style": "photorealistic"
}'
```

Example (Async):

```
curl -X POST https://api.greycats.com/api/ai/creative/image \
-H "Authorization: Bearer <TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "clientId": 12,
  "intent": "Flat illustration of team collaboration",
  "platform": "linkedin",
  "aspectRatio": "16:9",
  "style": "flat design",
  "mode": "async"
}'
```

Notes:

- **Two-step generation:** The user's **intent** is first transformed into a detailed, optimized image generation prompt by the text AI. This engineered prompt is then sent to the image generation API. The **promptUsed** field in the response shows the engineered prompt.
- **Sync mode** blocks until the image is generated (typically 15-45 seconds). Use for real-time preview.
- **Async mode** returns immediately and generates the image in the background. Poll the assets endpoint (**GET /api/ai/creative/assets/:clientId**) to check when the image appears. If generation fails, the asset placeholder is deleted.
- Brand context (including color palette) is injected into the image prompt for brand-consistent visuals.
- Images are saved to **/uploads/ai/creative/** on disk. The **imageUrl** is a relative path served by the static file middleware.
- Uses OpenAI's **gpt-image-1** model by default. Handles both **b64_json** (gpt-image-1) and URL-based (DALL-E 2/3) responses.

4.4 Get Creative Assets

Retrieves saved creative assets (captions and images) for a client with pagination and optional type filtering.

Endpoint: **GET /api/ai/creative/assets/:clientId**

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
clientId	number	Client ID

Query Parameters:

Parameter	Type	Required	Default	Description
type	string	No	--	Filter by type: "caption" or "image"
page	number	No	1	Page number (1-indexed)
limit	number	No	20	Items per page (max 50)

Success Response (200):

```
{
  "success": true,
  "data": {
    "assets": [
      {
        "id": "clxyz-asset-1...",
        "userId": 5,
        "clientId": 12,
        "type": "image",
        "platform": "instagram",
        "goal": null,
        "topic": "Modern office workspace",
        "promptUsed": "A professional, modern office workspace...",
        "imageUrl": "/uploads/ai/creative/img-1716825600000.png",
        "imagePath": "/app/uploads/ai/creative/img-1716825600000.png",
        "aspectRatio": "1:1",
        "style": "photorealistic",
        "captions": null,
        "createdAt": "2026-05-27T15:00:00.000Z",
        "updatedAt": "2026-05-27T15:00:00.000Z"
      },
      {
        "id": "clxyz-asset-2...",
        "userId": 5,
        "clientId": 12,
        "type": "caption",
        "platform": "instagram",
        "goal": "engagement",
        "topic": "Summer campaign launch",
        "promptUsed": null,
        "imageUrl": null,
        "imagePath": null,
        "aspectRatio": null,
        "style": null,
        "captions": [
          {
            "id": "caption_1",
            "text": "Summer is here and so is our biggest launch yet...",
            "hashtags": ["#Summer", "#NewLaunch"],
            "characterCount": 52,
            "tone": "CASUAL"
          }
        ]
      }
    ],
    "createdAt": "2026-05-26T12:00:00.000Z",
    "updatedAt": "2026-05-26T12:00:00.000Z"
  }
},
```

```
{
  "total": 15,
  "page": 1,
  "limit": 20
}
```

Example:

```
curl -X GET "https://api.greycats.com/api/ai/creative/assets/12?type=image&page=1&limit=10" \
-H "Authorization: Bearer <TOKEN>"
```

Notes:

- Assets are ordered by **createdAt** descending (newest first).
- For **type**: "caption" assets, the **captions** field contains the array of generated captions with their metadata.
- For **type**: "image" assets, the **imageUrl** field contains the image path. If **imageUrl** is **null**, the image is still generating (async mode) or generation failed.

4.5 Delete Creative Asset

Permanently deletes a creative asset. For image assets, also removes the physical image file from disk.

Endpoint: **DELETE** `/api/ai/creative/asset/:assetId`

Auth: Bearer JWT required

Path Parameters:

Parameter	Type	Description
assetId	string	Creative asset ID

Success Response (200):

```
{
  "success": true,
  "message": "Asset deleted"
}
```

Error Responses:

Status	Condition
404	Asset not found or not owned by user

Example:

```
curl -X DELETE https://api.greycats.com/api/ai/creative/asset/clxyz-asset-1 \
-H "Authorization: Bearer <TOKEN>"
```

Notes:

- For image assets, the physical file at `imagePath` is deleted from disk.
- This is a permanent, irreversible deletion.

Appendix: Complete Route Summary

Method	Path	Feature	Description
POST	/api/ai/brand/scrape	Brand	Scrape website for brand info
POST	/api/ai/brand/profile	Brand	Create/update brand profile
GET	/api/ai/brand/profile/:clientId	Brand	Get brand profile + knowledge
POST	/api/ai/brand/knowledge	Brand	Upload knowledge base file
GET	/api/ai/brand/knowledge/:clientId	Brand	List knowledge base files
DELETE	/api/ai/brand/knowledge/:knowledgeId	Brand	Delete knowledge base file
POST	/api/ai/chat/session	Chatbot	Create/resume chat session
POST	/api/ai/chat/message	Chatbot	Send message (SSE stream)
GET	/api/ai/chat/session/:sessionId/messages	Chatbot	Get chat history
GET	/api/ai/chat/sessions	Chatbot	List all sessions
DELETE	/api/ai/chat/session/:sessionId	Chatbot	Delete session + messages
POST	/api/ai/chat/summary	Chatbot	Generate report summary
PUT	/api/ai/chat/message/:messageId/feedback	Chatbot	Rate message (thumbs up/down)

Method	Path	Feature	Description
PUT	/api/ai/chat/message/:messageId/action/:actionIndex/apply	Chatbot	Mark action as applied
GET	/api/ai/chat/analytics	Chatbot	Get AI usage analytics
POST	/api/ai/calendar/generate	Calendar	Generate calendar (async)
GET	/api/ai/calendar/:calendarId	Calendar	Get calendar entries
GET	/api/ai/calendar/client/:clientId	Calendar	List calendars for client
PUT	/api/ai/calendar/entry/:entryId	Calendar	Update calendar entry
DELETE	/api/ai/calendar/:calendarId	Calendar	Delete calendar
POST	/api/ai/calendar/entry/from-chat	Calendar	Create entry from chatbot
POST	/api/ai/calendar/entry/:entryId/schedule	Calendar	Convert to scheduled post
POST	/api/ai/creative/caption	Creative	Generate captions
POST	/api/ai/creative/content	Creative	Generate article/script
POST	/api/ai/creative/image	Creative	Generate image
GET	/api/ai/creative/assets/:clientId	Creative	List creative assets
DELETE	/api/ai/creative/asset/:assetId	Creative	Delete creative asset